10TH CLASS MATHEMATICS — NEW SYLLABUS (PECTAA)

Exercise 3.3 — Addition & Subtraction of Matrices

Unit 3 | Solved Exercise | New Book (PECTAA)

Exercise 3.3 (Solved)

1. Which of the following matrices are conformable for addition and subtraction?

Two matrices can be added or subtracted only when they have the same order, i.e. the same number of rows and the same number of columns. So first write the order of every matrix, then make the groups.

Order of each matrix

$A = \begin{bmatrix} 3 & 4 \\ -2 & 1 \end{bmatrix}$ (2×2)

$B = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$ (2×1)

$C = \begin{bmatrix} 5 & 2 \end{bmatrix}$ (1×2)

$D = \begin{bmatrix} 1 & 7 \\ 2 & 5 \end{bmatrix}$ (2×2)

$E = \begin{bmatrix} 2 \end{bmatrix}$ (1×1)

$F = \begin{bmatrix} 7 & 11 \end{bmatrix}$ (1×2)

$G = \begin{bmatrix} a \\ b \end{bmatrix}$ (2×1)

$H = \begin{bmatrix} 3 \end{bmatrix}$ (1×1)

$M = \begin{bmatrix} l \\ m \end{bmatrix}$ (2×1)

Grouping matrices of the same order

A and D are both $2 \times 2 \implies$ conformable

C and F are both $1 \times 2 \implies$ conformable

E and H are both $1 \times 1 \implies$ conformable

G and M are both $2 \times 1 \implies$ conformable

B is 3×1, and no other matrix has order 3×1

$\therefore$ B is not conformable with any matrix given here

Answer: Conformable for addition / subtraction: A & D, C & F, E & H, G & M

2. If X , Y and Z are the following matrices, then find:

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$$X = \begin{bmatrix} 1 & -1 \\ -2 & 2 \end{bmatrix} \quad Y = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \quad Z = \begin{bmatrix} 3 & 0 \\ 0 & -2 \end{bmatrix}$$

(i) $X + Y$ (ii) $Y + 7Z$ (iii) $4X - Z$

(iv) $X + 2Y + 3Z$ (v) $X - 4Y + Z$ (vi) $Z - Z$

(i) X + Y

$$\begin{bmatrix} 1 & -1 \\ -2 & 2 \end{bmatrix} + \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 6 \end{bmatrix}$$

Add the corresponding elements

Answer: X + Y = $\begin{bmatrix} 2 & 1 \\ 1 & 6 \end{bmatrix}$

(ii) Y + 7Z

7Z = 7 × Z, so multiply every element of Z by 7

$$7Z = \begin{bmatrix} 21 & 0 \\ 0 & -14 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} + \begin{bmatrix} 21 & 0 \\ 0 & -14 \end{bmatrix} = \begin{bmatrix} 22 & 2 \\ 3 & -10 \end{bmatrix}$$

Answer: Y + 7Z = $\begin{bmatrix} 22 & 2 \\ 3 & -10 \end{bmatrix}$

(iii) 4X − Z

$$4X = \begin{bmatrix} 4 & -4 \\ -8 & 8 \end{bmatrix}$$

$$\begin{bmatrix} 4 & -4 \\ -8 & 8 \end{bmatrix} - \begin{bmatrix} 3 & 0 \\ 0 & -2 \end{bmatrix} = \begin{bmatrix} 1 & -4 \\ -8 & 10 \end{bmatrix}$$

Answer: 4X − Z = $\begin{bmatrix} 1 & -4 \\ -8 & 10 \end{bmatrix}$

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(iv) X + 2Y + 3Z

$$2Y = \begin{bmatrix} 2 & 4 \\ 6 & 8 \end{bmatrix} \qquad 3Z = \begin{bmatrix} 9 & 0 \\ 0 & -6 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -1 \\ -2 & 2 \end{bmatrix} + \begin{bmatrix} 2 & 4 \\ 6 & 8 \end{bmatrix} + \begin{bmatrix} 9 & 0 \\ 0 & -6 \end{bmatrix}$$

$$= \begin{bmatrix} 12 & 3 \\ 4 & 4 \end{bmatrix}$$

Answer: X + 2Y + 3Z = $\begin{bmatrix} 12 & 3 \\ 4 & 4 \end{bmatrix}$

(v) X − 4Y + Z

$$4Y = \begin{bmatrix} 4 & 8 \\ 12 & 16 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -1 \\ -2 & 2 \end{bmatrix} - \begin{bmatrix} 4 & 8 \\ 12 & 16 \end{bmatrix} + \begin{bmatrix} 3 & 0 \\ 0 & -2 \end{bmatrix}$$

$$= \begin{bmatrix} -3 & -9 \\ -14 & 0 \end{bmatrix} + \begin{bmatrix} 3 & 0 \\ 0 & -2 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & -9 \\ -14 & -2 \end{bmatrix}$$

Answer: X − 4Y + Z = $\begin{bmatrix} 0 & -9 \\ -14 & -2 \end{bmatrix}$

(vi) Z − Z

$$\begin{bmatrix} 3 & 0 \\ 0 & -2 \end{bmatrix} - \begin{bmatrix} 3 & 0 \\ 0 & -2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Every element becomes zero, so the result is the null matrix

Answer: Z − Z = $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = O$

3. Find the additive inverse of the following matrices:

The additive inverse of A is −A. We change the sign of every element of A, because A + (−A) = O.

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(i) $P = \begin{bmatrix} 5 \\ -7 \end{bmatrix}$

$-P = \begin{bmatrix} -5 \\ 7 \end{bmatrix}$

Check: $P + (-P) = O$

Answer: additive inverse = $\begin{bmatrix} -5 \\ 7 \end{bmatrix}$

(ii) $Q = \begin{bmatrix} 9 & -3 \end{bmatrix}$

$-Q = \begin{bmatrix} -9 & 3 \end{bmatrix}$

Answer: additive inverse = $\begin{bmatrix} -9 & 3 \end{bmatrix}$

(iii) $R = \begin{bmatrix} 2 & -1 \\ -2 & 3 \end{bmatrix}$

$-R = \begin{bmatrix} -2 & 1 \\ 2 & -3 \end{bmatrix}$

Answer: additive inverse = $\begin{bmatrix} -2 & 1 \\ 2 & -3 \end{bmatrix}$

(iv) $S = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ **(unit matrix)**

$-S = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$

Answer: additive inverse = $\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$

4. If A , B and C are the following matrices, then verify:

$A = \begin{bmatrix} 2 & 3 \\ -3 & 2 \end{bmatrix}$ $B = \begin{bmatrix} 3 & 4 \\ 5 & 6 \end{bmatrix}$ $C = \begin{bmatrix} 1 & -2 \\ 0 & 5 \end{bmatrix}$

(i) $A + B = B + A$

(ii) $(A + B) + C = A + (B + C)$

(iii) $(2A + B) + C = 2A + (B + C)$

(iv) $3(A + B) = 3A + 3B$

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(i) A + B = B + A

$$A + B = \begin{bmatrix} 2 & 3 \\ -3 & 2 \end{bmatrix} + \begin{bmatrix} 3 & 4 \\ 5 & 6 \end{bmatrix} = \begin{bmatrix} 5 & 7 \\ 2 & 8 \end{bmatrix}$$

$$B + A = \begin{bmatrix} 3 & 4 \\ 5 & 6 \end{bmatrix} + \begin{bmatrix} 2 & 3 \\ -3 & 2 \end{bmatrix} = \begin{bmatrix} 5 & 7 \\ 2 & 8 \end{bmatrix}$$

Both sides are equal, so addition of matrices is commutative

Answer: A + B = B + A (verified)

(ii) (A + B) + C = A + (B + C)

$$B + C = \begin{bmatrix} 3 & 4 \\ 5 & 6 \end{bmatrix} + \begin{bmatrix} 1 & -2 \\ 0 & 5 \end{bmatrix} = \begin{bmatrix} 4 & 2 \\ 5 & 11 \end{bmatrix}$$

$$A + B = \begin{bmatrix} 5 & 7 \\ 2 & 8 \end{bmatrix}$$

$$\text{L.H.S.} = (A + B) + C = \begin{bmatrix} 5 & 7 \\ 2 & 8 \end{bmatrix} + \begin{bmatrix} 1 & -2 \\ 0 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} 6 & 5 \\ 2 & 13 \end{bmatrix}$$

$$\text{R.H.S.} = A + (B + C) = \begin{bmatrix} 2 & 3 \\ -3 & 2 \end{bmatrix} + \begin{bmatrix} 4 & 2 \\ 5 & 11 \end{bmatrix}$$

$$= \begin{bmatrix} 6 & 5 \\ 2 & 13 \end{bmatrix}$$

Answer: L.H.S. = R.H.S. (verified)

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(iii) (2A + B) + C = 2A + (B + C)

$$2A = \begin{bmatrix} 4 & 6 \\ -6 & 4 \end{bmatrix}$$

$$2A + B = \begin{bmatrix} 4 & 6 \\ -6 & 4 \end{bmatrix} + \begin{bmatrix} 3 & 4 \\ 5 & 6 \end{bmatrix} = \begin{bmatrix} 7 & 10 \\ -1 & 10 \end{bmatrix}$$

$$\text{L.H.S.} = (2A + B) + C = \begin{bmatrix} 7 & 10 \\ -1 & 10 \end{bmatrix} + \begin{bmatrix} 1 & -2 \\ 0 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} 8 & 8 \\ -1 & 15 \end{bmatrix}$$

$$\text{R.H.S.} = 2A + (B + C) = \begin{bmatrix} 4 & 6 \\ -6 & 4 \end{bmatrix} + \begin{bmatrix} 4 & 2 \\ 5 & 11 \end{bmatrix}$$

$$= \begin{bmatrix} 8 & 8 \\ -1 & 15 \end{bmatrix}$$

Answer: L.H.S. = R.H.S. (verified)

(iv) 3(A + B) = 3A + 3B

$$\text{L.H.S.} = 3(A + B) = 3 \begin{bmatrix} 5 & 7 \\ 2 & 8 \end{bmatrix} = \begin{bmatrix} 15 & 21 \\ 6 & 24 \end{bmatrix}$$

$$3A = \begin{bmatrix} 6 & 9 \\ -9 & 6 \end{bmatrix} \quad 3B = \begin{bmatrix} 9 & 12 \\ 15 & 18 \end{bmatrix}$$

$$\text{R.H.S.} = 3A + 3B = \begin{bmatrix} 6 & 9 \\ -9 & 6 \end{bmatrix} + \begin{bmatrix} 9 & 12 \\ 15 & 18 \end{bmatrix}$$

$$= \begin{bmatrix} 15 & 21 \\ 6 & 24 \end{bmatrix}$$

Answer: L.H.S. = R.H.S. (verified)

5. If A and B are the following matrices, then show that B is additive inverse of A and A is additive inverse of B.

$$A = \begin{bmatrix} 5 & 6 \\ 7 & -2 \end{bmatrix} \quad B = \begin{bmatrix} -5 & -6 \\ -7 & 2 \end{bmatrix}$$

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Solution

$$A + B = \begin{bmatrix} 5 & 6 \\ 7 & -2 \end{bmatrix} + \begin{bmatrix} -5 & -6 \\ -7 & 2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = O$$

$$B + A = \begin{bmatrix} -5 & -6 \\ -7 & 2 \end{bmatrix} + \begin{bmatrix} 5 & 6 \\ 7 & -2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = O$$

In both orders the sum is the null matrix O

Answer: Hence A and B are additive inverse of each other (proved)

6. If A and B are the following matrices, then verify that:

$$A = \begin{bmatrix} 6 & -2 \\ 0 & 3 \end{bmatrix} \quad B = \begin{bmatrix} 8 & 1 \\ 3 & 0 \end{bmatrix}$$

(i) $(A + B)^t = A^t + B^t$ (ii) $(A - B)^t = A^t - B^t$

(i) $(A + B)^t = A^t + B^t$

$$A + B = \begin{bmatrix} 6 & -2 \\ 0 & 3 \end{bmatrix} + \begin{bmatrix} 8 & 1 \\ 3 & 0 \end{bmatrix} = \begin{bmatrix} 14 & -1 \\ 3 & 3 \end{bmatrix}$$

L.H.S. = (A + B)^{t} = $\begin{bmatrix} 14 & 3 \\ -1 & 3 \end{bmatrix}$

A^{t} = $\begin{bmatrix} 6 & 0 \\ -2 & 3 \end{bmatrix}$ B^{t} = $\begin{bmatrix} 8 & 3 \\ 1 & 0 \end{bmatrix}$

R.H.S. = A^{t} + B^{t} = $\begin{bmatrix} 14 & 3 \\ -1 & 3 \end{bmatrix}$

Answer: L.H.S. = R.H.S. (verified)

(ii) $(A - B)^t = A^t - B^t$

$$A - B = \begin{bmatrix} 6 & -2 \\ 0 & 3 \end{bmatrix} - \begin{bmatrix} 8 & 1 \\ 3 & 0 \end{bmatrix} = \begin{bmatrix} -2 & -3 \\ -3 & 3 \end{bmatrix}$$

L.H.S. = (A − B)^{t} = $\begin{bmatrix} -2 & -3 \\ -3 & 3 \end{bmatrix}$

R.H.S. = A^{t} − B^{t} = $\begin{bmatrix} 6 & 0 \\ -2 & 3 \end{bmatrix} - \begin{bmatrix} 8 & 3 \\ 1 & 0 \end{bmatrix}$

= $\begin{bmatrix} -2 & -3 \\ -3 & 3 \end{bmatrix}$

Answer: L.H.S. = R.H.S. (verified)

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